

JOSE E. RODRÍGUEZ-RÍOS

COMPUTER SCIENTIST | AI/HPC | CYBERSECURITY

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PROFESSIONAL PROFILE

Computer scientist and cybersecurity engineer whose work connects AI, high-performance computing, secure research software, and network defense. National-laboratory experience spans regulatory-genomics training on leadership-class systems at Argonne and containerized cyber-research infrastructure at Lawrence Livermore; community work includes an energy-billing application for a solar cooperative in Puerto Rico.

Research interests: AI for scientific computing; research software engineering; secure distributed systems; vulnerability assessment; cybersecurity education

EDUCATION

Ph.D., Software Engineering | University of Puerto Rico - Mayagüez

Aug 2026 - Present

M.S., Computer Science | University of Puerto Rico - Río Piedras

Aug 2022 - May 2025 | GPA: 4.0/4.0

B.S., Computer Science | University of Puerto Rico - Río Piedras

Aug 2015 - Dec 2020 | Minor: Cybersecurity

TECHNICAL SKILLS

Programming: Python, C, C++, Java, Assembly

AI/ML & Data: PyTorch, TensorFlow, Keras, Hugging Face, LLMs/transformers, Pandas, NumPy, Matplotlib, Jupyter

High-Performance Computing: SLURM, GPU computing, NERSC Perlmutter, ALCF Polaris, ALCF Aurora, Jupyter-to-batch workflows

Cybersecurity & Forensics: Vulnerability assessment, reverse engineering, digital forensics, Nmap, OpenVAS, Wireshark, Ghidra, Autopsy, Scapy; trained on Metasploit, Hashcat, Burp Suite

DevOps & Platforms: Linux, Docker, GitLab CI/CD and runners, Git/GitHub, Elasticsearch, Ollama; familiar with Kubernetes and OpenShift

Networking & VPN: OpenVPN, WireGuard, client-server deployments

Web, Data & Automation: Flask, Dash, Streamlit, Playwright, MySQL, SQLite, openpyxl

Operating Systems: Linux, Windows, Kali Linux

Languages: English, Spanish

INDUSTRY AND NATIONAL LABORATORY EXPERIENCE

Research Intern | Argonne National Laboratory | Lemont, IL

Jun 2026 - Aug 2026

Project: Leveraging Large Language Models for Regulatory Genomics on HPC

- Developed and tested an educational DNA sequence-classification workflow with PyTorch, Hugging Face, and DNABERT-style transformers on NERSC Perlmutter and ALCF Polaris and Aurora, spanning CPU baselines and GPU-accelerated training.
- Compared 6-mer and byte-pair-encoding (BPE) tokenization for DNA: similar accuracy (81% vs. 80%) while BPE cut tokens per sequence from 195 to 34.
- Designed fixed-configuration CPU-versus-GPU experiments to isolate acceleration benefits, then scaled workloads to demonstrate GPU-enabled capability.
- Created Jupyter lessons and diagrams covering transformer architectures, embeddings, classification pipelines, and SLURM commands including sbatch, squeue, and sacct.
- Led a 10-student team within the 87-student bootcamp as project lead: scoped the workflow to a 10-12-hour project and coached two sub-teams through their final presentations to peers and laboratory staff.
- Guided one sub-team to train a DNA transformer from scratch, compare five tokenizations, and reach a validation AUROC of 0.887 with overlapping 6-mers.
- Guided the second sub-team to fine-tune DNABERT as a CTCF-binding classifier on K562 data, reaching 91.7% evaluation accuracy (99.1% training) after about 17 minutes of training, with external testing on an astrocyte CTCF dataset.
- Authored a first-time project-lead guide and reusable project template to onboard future interns.

DevOps Intern | *Lawrence Livermore National Laboratory* | Livermore, CA

Jun 2024 - Aug 2024

- Containerized cyber-research applications and integrated GitLab-based CI/CD pipelines into cyber research workflows in an Agile environment.
- Built an Elasticsearch container for Ghidra BSim-extracted binary features, enabling security researchers to query malware-like samples at scale.
- Deployed a Docker-hosted Ollama (Llama 3) large language model service for the PaperQA PDF-reading application over LLNL research papers.

Cybersecurity Intern | *GM Sectech* | San Juan, PR

Jun 2022 - Aug 2022

- Collaborated with penetration testers on credential-theft, password-cracking, and domain-compromise scenarios; trained on Metasploit, Hashcat, and Burp Suite.
- Researched and executed internal attack techniques including phishing and malicious USB delivery for awareness training.
- Developed a Python/Twilio smishing (SMS phishing) simulator that sent controlled SMS lures and captured user interaction for security education.

RESEARCH AND ACADEMIC EXPERIENCE

Graduate Research and Teaching Assistant

Aug 2022 - May 2025

Laboratory of Dr. José Ortiz Ubarri and Department of Computer Science, UPR Río Piedras | San Juan, PR

- Designed and implemented NVSRCO, a network situational-awareness and vulnerability-scanning platform for organizations with limited security budgets, built entirely on open-source components so the only cost is the analysis server.
- Combined host discovery, port and service enumeration, vulnerability assessment, persistent scan storage, and secure web reporting in an end-to-end system that scans every host on a network; each vulnerability scan takes about 30 minutes.
- Integrated Python, Nmap, and OpenVAS into a remote scanning workflow. The published implementation uses CSV reports, a Dash dashboard, WireGuard connectivity, and Docker configurations.
- Led laboratory sessions for Operating Systems, Computer Architecture, and Cybersecurity courses, training undergraduates in Linux, Ghidra-based assembly inspection, buffer overflow discovery, exploit development, and packet analysis with Wireshark and Scapy.
- Authored Capture the Flag (CTF) challenges in reverse engineering, cryptography, web, and forensics with progressive difficulty, and ran coding and outreach events for about 60 high-school students from Puerto Rico.

Undergraduate Research Assistant

2020

Laboratory of Dr. José Ortiz Ubarri, UPR Río Piedras | San Juan, PR

- Developed a deception-based defense project: a decoy university website using honeypot techniques that drew about 1,000 automated attack attempts per day and tracked where the attacks originated.

Undergraduate Software Assistant

2019

Molecular Biology Laboratory of Dr. José A. Rodríguez-Martínez, UPR Río Piedras | San Juan, PR

- Maintained and extended NRLB, a Java tool for analyzing large-scale protein-DNA interaction data from sequencing experiments supported by NIH SC1GM127231.
- Adapted analysis code for specific DNA-binding sequences and provided software support to laboratory researchers.

COMMUNITY AND LEADERSHIP

Volunteer Software Developer | *Cooperativa de Energía La Margarita (Abeyno Coop)* | Salinas, PR

2024 - Present

- Built Solar Fleet Reporter, a local Python/Streamlit application (about 1,700 lines, 88 offline test checks) that pulls monthly energy data for the cooperative's solar fleet from Sol-Ark, FranklinWH, and sonnen portals through their APIs and Playwright-driven logins.
- Implemented the cooperative's net-metering billing rules (three billing cases, per-site rates), CSV and Excel export, printable homeowner statements, and offline-site alerts so members can be billed accurately each month.

Peer Mentor | *Intro to HPC Bootcamp, Argonne National Laboratory*

Aug 2025

- Mentored a 5-student team (one of two peer mentors on a 10-student project) applying Python and data science to design an energy-efficient HPC system for fusion workloads under a 12 MW power cap.

SELECTED PROJECTS

Network Vulnerability and Situational Awareness Scanner (NVSRCO)

Aug 2022 - Dec 2025

Graduate research project

Project: Source and report | github.com/joserico00/NVSRCO | repositorio.upr.edu/handle/11721/4342

- Scanner built on open-source components and described under Research and Academic Experience; the technical report is published in the UPR institutional repository.

Deep Learning for Malicious Traffic Detection | Coursework and continued development

2024 - 2026

- Developed TensorFlow/Keras experiments on IoT-23 network-connection logs; the current pipeline splits by capture and fits preprocessing only on training data to prevent leakage.
- Evaluates a class-weighted dense neural network against a logistic-regression baseline using balanced accuracy, precision, recall, F1, ROC AUC, and PR AUC; earlier model runs and their limitations are documented separately.

Malware Image Classification (Maling) | Deep-learning project

- Trained a TensorFlow/Keras convolutional neural network on the Maling dataset (9,401 grayscale malware images, 25 families) to 97.7% validation accuracy.

Operating Systems Projects | UPR Río Piedras

2022

Project: Source | github.com/joserico00/Operating-System-Projects

- Implemented FIFO, Working Set Clock, and Optimal page-replacement algorithms over a simulated virtual-memory list in Python.
- Modeled producer-consumer coordination with sockets, threads, and mutex-protected critical sections.
- Built a distributed file system with metadata and datanode servers supporting list and copy operations over sockets.

University Enrollment Database | UPR Río Piedras

2019

- Developed a normalized MySQL database modeling course enrollment, professors, courses, and students with unique identifiers.

ALCF Compute Finder

2026

- Built a Streamlit workload-to-system recommendation tool with explainable rankings, hard-requirement checks, job-script guidance, and a browser-based stlite option; unknown specifications never satisfy a requirement.

1UPPER Game Catalog Enrichment Pipeline

2026

- Built resumable Python ingestion and enrichment scripts for a software-engineering project, combining IGDB metadata and playtime information into CSVs for database import; implemented retries, checkpoints, caching, and offline tests.

Reporter Assay Analysis

2026

- Built a notebook workflow that reshapes luciferase-assay spreadsheets into tidy data, compares conditions against a baseline, and produces figures; included a synthetic example workbook.

Epidemic Modeling Teaching Notebooks

2026

- Developed NumPy teaching workflows for deterministic SIR models and stochastic outbreak simulations, with two student tracks, a master notebook, reference metrics, and HTML previews.

yt2ogg Audio Loop Tool

2026

- Built a shared desktop/CLI pipeline for seamless Ogg loops using NumPy beat tracking, FFmpeg audio processing, and standalone executable packaging with self-tests.

WORKSHOPS AND OUTREACH

- Intro to HPC Bootcamp Participant | Berkeley Lab / NERSC** **2023**
- Worked with a team on heatwave and power-outage analysis using Python, pandas, Matplotlib, and Perlmutter; presented results in a lightning talk. Power-outage notebooks.
- Cybersecurity Workshop Developer | UPR Río Piedras** **2023 - 2025**
- Created reusable CTF challenge generators and walkthroughs covering cryptography, forensics, Linux, scripting, and related security topics. CTF workshop materials.
- Volunteer | Argonne 80th Anniversary Open House** **2026**
- Helped run a human neural-network demonstration and built an LLM-assisted version to support single-visitor participation.

COMPETITIONS AND HONORS

- 1st Place (21 teams) - Data Analytics Challenge | CAHSI** **2026**
- Built a LightGBM network-intrusion classifier, achieving 0.9428 Macro-F1. Classified benign traffic and 35 attack types using approximately 1.7 million training flows, with leakage-safe validation and class-imbalance handling.
- 2nd Place, Advanced Team Category - Cyber Hackathon | CAHSI / GMIS** **2026**
- Solved penetration-testing, reverse-engineering, and digital-forensics challenges in a jeopardy-style Capture the Flag competition.
- 1st Place (30 teams) - Capture the Flag (CTF) | BSides Puerto Rico | San Juan, PR** **Apr 2024**
- Solved challenges in reverse engineering, cryptography, network analysis, and binary exploitation as part of a 3-member team.
- Selected Student (1 of 29 from 400+) - HSI Tech Summit | Bloomberg | New York, NY** **Apr 2024**
- Selected from 400+ applicants for live coding and engineering sessions; later invited to the 6-week Bloomberg Summer Accelerator.
- 5th Place - Bloomberg Hackathon | UPR Río Piedras | San Juan, PR** **Sep 2022**
- Developed Seniors, a Python/PySimpleGUI/SQLite prototype connecting older adults with nearby caregivers or volunteers through location-based requests.
- 2nd Place (team, 40 participants) - Tracer Fire Blue Team Competition | ACSAC (Annual Computer Security Applications Conference)** **Dec 2019**
- Diagnosed compromised accounts and services with SIEM and IDS tools; performed disk and memory forensics with Autopsy, malware analysis with Wireshark, and binary reversing with Ghidra.

SCHOLARSHIPS

- Francis Castro Scholarship, University of Puerto Rico (2020, 2022)
- Puerto Rico Louis Stokes Alliance for Minority Participation (PR-LSAMP) Undergraduate Fellow (2020)

REFERENCES

Available upon request.